

Zeyu Wang

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EDUCATION

Sun Yat-sen University, China

Sep 2021 - Jun 2024

Master in Electronic and Information Engineering

GPA: 3.60/4.0

- Research Interests: Image Processing; Deep Learning; LCD Backlight; Display Technologies
- Scholarship: First-Class Scholarship

Admitted to Master's program via recommendation, exempt from entrance exam

China University of Mining and Technology, China

Sep 2017 - Jun 2021

Bachelor in Electronic and Information Engineering

GPA: 3.61/4.0

- Scholarship: First-Class Scholarship

PUBLICATIONS

Journals

- **Z. Wang**, G. Zou, Y. Shen, B.R. Yang, Z. Qin. Deep learning-based real-time driving for 3-field sequential color displays with low color breakup and high fidelity. *Optics Express*, 2023, doi: 10.1364/OE.487198
- **Z. Wang**, G. Zou, Y. Li, C. Ma, Z. Zhang, G. Zha, Y. Shen, B.R. Yang, Z. Qin. Two-field sequential color liquid crystal displays with deep learning-enabled real-time driving. *Optics Letters*, 2023, doi: 10.1364/OL.501567
- G. Zou, **Z. Wang**, Y. Liu, J. Li, X. Liu, J. Liu, B.R. Yang, Z. Qin. Deep learning-enabled image content-adaptive field sequential color LCDs with mini-LED backlight. *Optics Express*, 2022, doi: 10.1364/OE.459752

Conferences

- **Z. Wang**, G. Zou, Y. Shen, B.R. Yang, Z. Qin. Deep Learning Designed Driving Method for Field Sequential Color Displays. *Frontiers in Optics*, Rochester, NY, USA, 2022, doi: 10.1364/FIO.2022.JW4B.49
- **Z. Wang**, G. Zou, B.R. Yang, Z. Qin. 75-3: Field sequential color displays with simultaneously suppressed color breakup and flicker based on multi-objective optimization. *SID Symposium Digest of Technical Papers*, San Jose, CA, USA, 2022, doi: 10.1002/sdtp.15670
- Z. Qin, **Z. Wang**, Q. Wang, G. Zou, Y. Cheng, Y. Liu, Y. Li, C. Ma, Z. Zhang. High-resolution light field display based on a mini-LED field sequential color micro-LCD. *SPIE Photonics West*, San Francisco, CA, USA, 2024, doi: 10.1117/12.2692767 (Invited talk)

Patents

- Z. Qin, G. Zou, **Z. Wang**, H. Liang, B.R. Yang. Adaptive Refresh Rate Field Sequential Color Display Driving Method, Apparatus and Related Device. Chinese Invention Patent (Granted), CN202210380314.3, 2022.

Textbook

- *Wearable Optoelectronic Display Technology*, Science Press, Beijing, China, 2022.
Primary author of Chapter 2, focusing on photometry, colorimetry, and display system performance evaluation.

INDUSTRIAL PROJECTS

Field Sequential Color Display for Pancake VR System

Jan 2023 - Dec 2023

Project Leader | In collaboration with China Star Optoelectronics Technology (CSOT)

- Developed a driving method integrating **multi-objective optimization** and **deep learning** to enable real-time high-resolution image generation with minimal distortion and color breakup.

- Applied the driving method to a Pancake VR prototype featuring dual 2K displays on an FPGA platform, achieving **real-time performance at 60 FPS**.
- Demonstrated the VR prototype at CES 2024, Las Vegas, NV, USA.

RESEARCH PROJECTS

Developed Driving method for LCoS-based AR-HUD System

Jun 2021 - Feb 2022

Algorithm Development Leader

- Developed an **ultra-low-complexity driving method** for the system, leveraging **global dimming** to suppress color breakup while preserving image quality.
- Enabled real-time generation of high-resolution images with high brightness and optical efficiency for automotive AR-HUD applications.

WORK EXPERIENCE

Tencent, Shenzhen, China

Jun 2024 - Nov 2025

Semiconductor supply chain management

- Managed self-developed server components and developed supply plans aligned with business requirements and market conditions.
- Coordinated cross-functional teams to improve supply chain efficiency and reduce delays through proactive coordination and data-driven planning.
- Participated in supplier evaluation and onboarding for in-house products, and handled production scheduling and lower-tier material management.

SKILLS & INTERESTS

Skills: Python, C++, OpenCV, Matlab, Lighttools

Languages: Mandarin, English (IELTS 7)

Interests: Photography, Traveling, Hiking